Roadmap

Week 1 — Absolute Foundations

Goal: Learn how images, video streams, and colors work in OpenCV, and confidently perform basic image manipulation.

Day 1: Understanding the basics

- Be able to capture video from a camera
- Understand what a frame is
- Know how pixels and arrays represent images
- Display frames in color and in grayscale

Day 2: Understanding Video Streams in OpenCV

- Understand how frames are read from a camera
- Know what ret and frame mean
- Understand how OpenCV creates live video using frames
- Clearly know why while True is mandatory

Day 3: Controlling Playback & Exiting Video

- Understand how keyboard input is captured in OpenCV
- Know what cv2.waitKey() does and what it returns
- Understand ASCII values and why keys are represented as numbers
- Understand bitwise AND (&) and why 0xFF is used
- Implement a keyboard-controlled exit from a video loop

Day 4: Images as NumPy Arrays

- Understand images as NumPy arrays in OpenCV
- Read and display images using OpenCV
- Understand image shape, channels, and pixel values
- Differentiate between grayscale and color images

Day 5: Drawing on Images with OpenCV

- Draw **lines, rectangles, circles, and text** on images
- Control **color, thickness, and position** of drawings
- Modify an image array directly

Day 6: Image Resizing, Cropping, and Copying

- Resize images using OpenCV
- Crop images using array slicing
- Copy and modify image regions
- Understand how image dimensions change after resizing and cropping

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Day 7: Image Color Spaces

- Understand what a color space is
- Know the difference between BGR, Grayscale, and HSV
- Convert images between color spaces using OpenCV
- Understand when and why different color spaces are used

◇ Week 2 – Color-Based Detection to Tracking

Goal: Understand and build a complete real-time color-based object tracking pipeline, starting from HSV color detection and ending with a stable, polished tracker.

Day 8: Color Detection using HSV

- Detect a specific color in an image
- Use HSV ranges to isolate colors
- Create and apply masks
- Understand how color-based segmentation works

Day 9: Trackbars for real-time HSV tuning

- Create trackbars in OpenCV
- Adjust HSV values in real time
- Tune color detection interactively
- Understand how real-time parameter control works

Day 10: Color Detection in Live Video

- Capture live video from a webcam
- Apply HSV-based color detection on video frames
- Combine trackbars + video feed
- Perform real-time color segmentation

Day 11: Noise Removal & Morphological Operations

- Understand what **noise** is in images and masks
- Clean binary masks using **morphological operations**
- Use **erosion, dilation, opening, and closing**
- Improve color detection results

Day 12: Contour Detection

- Understand what contours are
- Detect contours from a binary image
- Filter contours by area
- Draw bounding boxes around detected objects

Day 13: Object Tracking Basics

- Track a colored object across video frames
- Use contours to find object position
- Identify the **largest object** of a given color
- Draw tracking visuals (center point, path)

Day 14: Polishing the Color-Based Object Tracker

- Make the tracker **more stable**
- Reduce jitter and sudden jumps
- Visualize motion clearly
- Improve usability and presentation

◇ Week 3

Goal: Learn how professional tracking algorithms follow objects across frames and understand when to use detection versus tracking.

Day 15: Background Subtraction & Motion-Based Tracking

- Understand motion detection vs color-based detection
- Learn how background subtraction works conceptually
- Detect moving objects without HSV
- Know when background subtraction is the right tool

Day 16: Motion Mask Cleanup & Reliability

- Understand why raw motion masks are unreliable
- Learn how noise, holes, and shadows appear in motion detection
- Know how masks are cleaned before contour detection
- Be able to reason about stable vs accurate detection

Day 17: Hybrid Tracking (Motion + Color)

- Understand why single-method tracking is unreliable
- Learn the concept of combining motion detection with color detection
- Know how motion limits the search area
- Understand how hybrid tracking improves accuracy and stability

Day 18: Introduction to Object Tracking Algorithms

- Understand the difference between object detection and object tracking.
- Learn why tracking algorithms are needed after detection.
- Understand the concept of initializing a tracker with a Region of Interest (ROI).
- Know the strengths and limitations of common OpenCV trackers (KCF, CSRT, MOSSE, etc.).

Day 19: Tracking Objects with OpenCV Trackers

- Initialize an OpenCV object tracker using a selected ROI.
- Track an object continuously across live video frames.
- Draw bounding boxes and tracking status on the video.
- Handle tracking success and tracking failure gracefully.

Day 20: Comparing Different Tracking Algorithms

- Understand the differences between CSRT, KCF, MOSSE, and MedianFlow.
- Compare trackers based on accuracy, speed, and robustness.
- Learn how different trackers behave under occlusion and fast motion.
- Know how to choose the appropriate tracker for different applications.

Day 21: Building a Complete Tracking Application

- Combine detection and tracking into a complete pipeline.
- Reinitialize tracking when the tracker loses the object.
- Visualize object trajectories and tracking information.
- Understand how real-world tracking systems combine multiple computer vision techniques.

◇ Week 4

Goal:

Day 22:

Day 23:

Day 24:

Day 25:

Day 26:

Day 27:

Day 28:

Day 29:

Day 30: